AI-BASED HEART DISEASE PREDICTION SYSTEM

Model Evaluation and Reliability Testing

Objective

Evaluate the trained classification model using cross-validation, a confusion matrix, ROC curve and AUC score. Compare training and validation performance to identify possible overfitting.

Evaluation Method

The planned evaluation uses stratified 5-fold cross-validation to assess consistency across folds. The confusion matrix measures correct and incorrect class predictions. The ROC curve plots the true-positive rate against the false-positive rate at different thresholds, and AUC summarizes the model's ability to distinguish between the two classes. Overfitting is assessed by comparing training, cross-validation and held-out test performance. Recorded Results

Metric	Recorded result
Test accuracy	88.89%
Recall / sensitivity	92.00%
ROC-AUC	0.96
Test samples	54

Confusion Matrix

Actual / Predicted	No disease	Disease
No disease	TN = 25	FP = 4
Disease	FN = 2	TP = 23

Interpretation: The recorded matrix has 2 false negatives and 4 false positives. False negatives are important to examine because they represent disease-positive cases the model missed.

Cross-validation and Overfitting Status

Cross-validation fold scores and the training-set score were not available for verification when this report was prepared. Add the five fold results and their mean ± standard deviation after running the current evaluation code. Compare those results with the held-out test results before concluding whether overfitting is present.

Conclusion

The recorded results are a promising initial indication of model performance, but reliability and overfitting cannot be confirmed until the cross-validation results and training-versus-test comparison are generated from the current code. The model is a prototype for clinical decision support and must not be used as a stand-alone diagnosis.